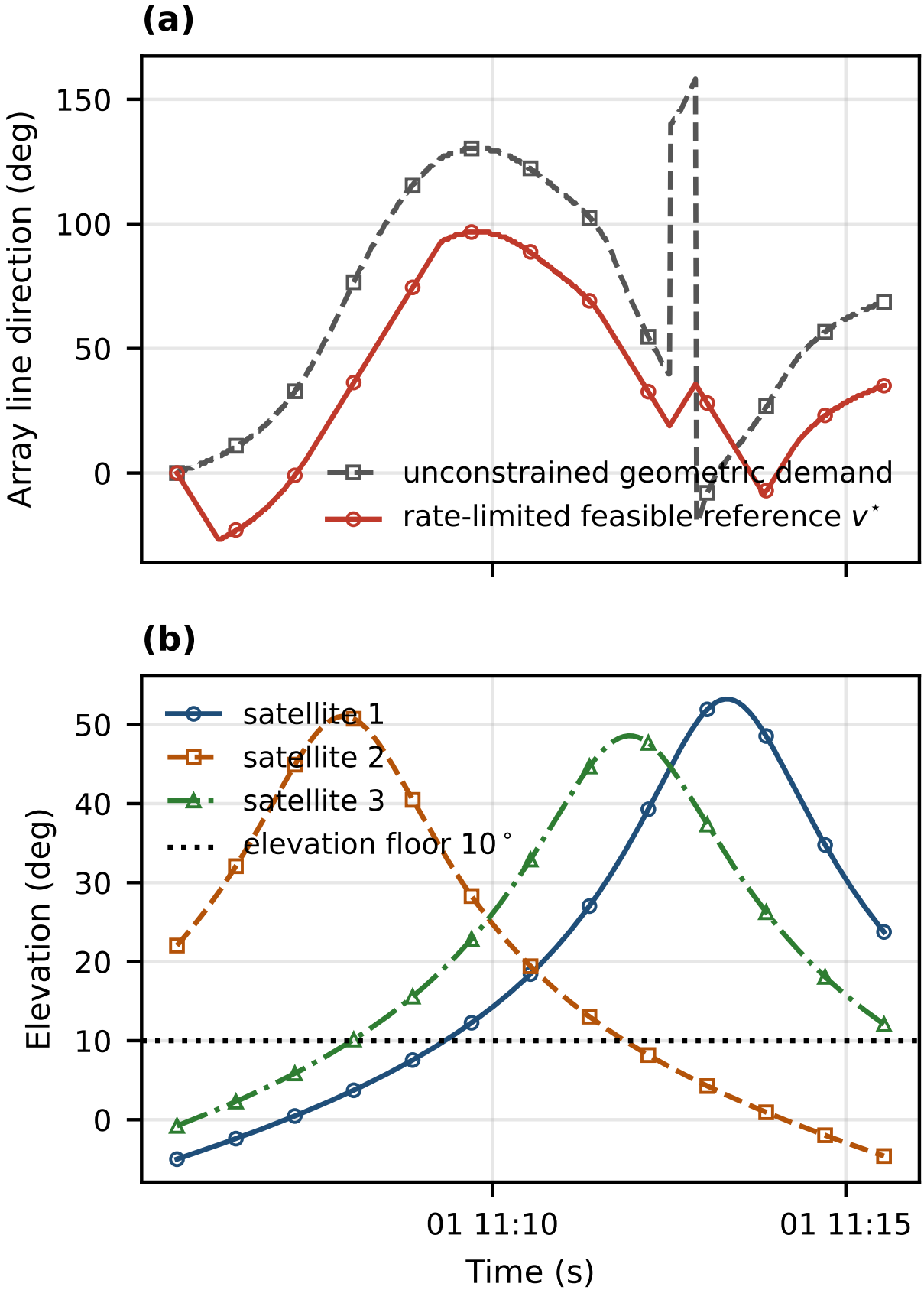k geometry: rate-limited reference direction and sate